

Gaussian Maximum-Likelihood Estimation: Closed Form, Gradient Descent, SGD, and Autograd

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Abstract

The Bernoulli program (`bernoulli/`) repeated for the 1D normal distribution: derive the closed-form MLE of (μ, σ) , reach the same optimum by gradient descent and by SGD on the negative log-likelihood, and verify that PyTorch autograd reproduces the hand-derived gradients to floating-point precision. New here: a two-dimensional parameter, and the fact that the MLE of the variance is the *biased* $1/n$ estimator.

1 Setup: from the likelihood to the NLL

Let $x_1, \dots, x_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with density $f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x - \mu)^2/2\sigma^2)$. By independence,

$$L(\mu, \sigma) = \prod_{i=1}^n f(x_i) = (2\pi)^{-n/2} \sigma^{-n} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right). \quad (1)$$

Taking log (strictly increasing: same maximizer), dividing by $n > 0$ and negating (neither moves the optimizer — the same three steps as in the Bernoulli note) gives the average negative log-likelihood

$$\mathcal{L}(\mu, \sigma) = \frac{1}{2} \log(2\pi) + \log \sigma + \frac{1}{2\sigma^2} \cdot \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2. \quad (2)$$

Expanding $(x_i - \mu)^2$ shows the data enters only through $\sum_i x_i$ and $\sum_i x_i^2$: a two-dimensional sufficient statistic.

2 Route 1: the theoretical solution

Proposition 1. *The unique MLE is*

$$\boxed{\hat{\mu} = \bar{x}, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.} \quad (3)$$

Proof. Differentiate (2) in μ :

$$\frac{\partial \mathcal{L}}{\partial \mu} = \frac{1}{\sigma^2} \cdot \frac{1}{n} \sum_i (\mu - x_i) = \frac{\mu - \bar{x}}{\sigma^2} = 0 \implies \mu = \bar{x}$$

(for every σ , so the two equations decouple). Differentiate in σ at $\mu = \bar{x}$, writing $v = \frac{1}{n} \sum_i (x_i - \bar{x})^2$:

$$\frac{\partial \mathcal{L}}{\partial \sigma} = \frac{1}{\sigma} - \frac{v}{\sigma^3} = 0 \implies \sigma^2 = v.$$

For uniqueness, $\mathcal{L}(\bar{x}, \sigma)$ has derivative $(\sigma^2 - v)/\sigma^3$: negative for $\sigma^2 < v$, positive for $\sigma^2 > v$, so v is the unique minimum in σ ; and for each σ the map $\mu \mapsto \mathcal{L}$ is a strictly convex parabola minimized at $\bar{x}$. $\square$

The $1/n$ is not a mistake. $\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n} \sigma^2$: the MLE of the variance is *biased*. Bessel's $\frac{1}{n-1}$ correction is a different estimator, chosen to be unbiased; maximum likelihood genuinely produces $1/n$. (The script prints both.)

3 Route 2: gradient descent

μ is already unconstrained; $\sigma > 0$ is not. Exactly as Bernoulli's $p = \text{sigmoid}(t)$, we evaluate the *same* NLL in an unconstrained coordinate, $\sigma = e^s$:

$$F(\mu, s) = \mathcal{L}(\mu, e^s) = \frac{1}{2} \log(2\pi) + s + \frac{1}{2e^{2s}} \cdot \frac{1}{n} \sum_i (x_i - \mu)^2. \quad (4)$$

Chain rule ($\partial \sigma / \partial s = e^s$):

$$\frac{\partial F}{\partial \mu} = \frac{\mu - \bar{x}}{e^{2s}}, \quad \frac{\partial F}{\partial s} = 1 - \frac{1}{e^{2s}} \cdot \frac{1}{n} \sum_i (x_i - \mu)^2. \quad (5)$$

Setting both to zero reproduces (3) — a consistency check on the derivation. GD iterates the full gradient $(\mu, s) \leftarrow (\mu, s) - \eta \nabla F$ and converges to it deterministically (both coordinates visible in the middle panel of the script's figure).

4 Route 3: stochastic gradient descent

Replace the full gradient with the one-sample estimate. Per sample i , $F_i(\mu, s) = s + (x_i - \mu)^2 / (2e^{2s}) + \text{const}$, so

$$g_i = \left(\frac{\mu - x_i}{e^{2s}}, \quad 1 - \frac{(x_i - \mu)^2}{e^{2s}} \right), \quad \mathbb{E}_i[g_i] = \nabla F, \quad (6)$$

an unbiased but noisy direction. With a constant step the iterate hovers in a noise ball around (3); averaging the final pass's iterates (Polyak–Ruppert) cancels the hovering. Mini-batches would average g_i over a batch — same expectation, variance $\div |B|$; engineering, not new mathematics.

5 Route 4: SGD with autograd

The script writes the per-sample loss *verbatim* as $s + (x_i - \mu)^2 / (2e^{2s})$ in torch and lets `backward()` differentiate the composition: it must produce exactly (6). Driving routes 3 and 4 through the identical sample schedule, step size, and initialization, the asserted bound $\max_k \|\theta_k^{\text{hand}} - \theta_k^{\text{autograd}}\| < 10^{-10}$ passes — autograd mechanizes the chain rule; it does not change the mathematics.

6 Experiment

With $(\mu, \sigma) = (2, 1.5)$ and $n = 5000$ (seed 7): the closed form gives $(\hat{\mu}, \hat{\sigma}) \approx (2.011, 1.502)$; GD converges to it in a few hundred steps; SGD's Polyak average lands within a few 10^{-2} (the variance coordinate is noisier than the mean's); and the hand and autograd trajectories coincide exactly. Same story as Bernoulli, one dimension higher: the likelihood surface decides, the optimizer only travels.